

A Phase-Field Dataset for Fracture in a Heterogeneous Concrete-Inspired Composite with Varying Notch Heights and Aggregate Positions

Zhan Liu^a, Abhinav Gupta^b, Somdatta Goswami^a

^a*Department of Civil and Systems Engineering, Johns Hopkins University, USA*

^b*Department of Civil and Environmental Engineering, Vanderbilt University, USA*

**This dataset was generated as part of NSF funded project
(2438193 – EAGER: NAIRR Pilot: Enabling parametric sweeps on exascale AI-integrated simulations
through federated learning)**

1. Introduction

Concrete and other cementitious composites fail in a quasi-brittle manner that is governed less by the applied load alone than by the internal mesostructure of the material. At the mesoscale, concrete is a two-phase medium in which stiff aggregates are embedded in a weaker cementitious matrix. When a crack propagates through such a medium, it does not follow the straight path predicted for a homogeneous solid: it is attracted to compliant regions, deflected around stiff inclusions, and can be temporarily arrested when it encounters aggregate before re-nucleating elsewhere. Predicting the crack path, and the peak load the specimen sustains along it, requires resolving the interaction between the initial defect and the surrounding heterogeneity. However, classical energy-based criteria of brittle fracture [1, 2] resolve this interaction only for idealized homogeneous geometries. This is the central difficulty in the computational modeling of concrete failure, making it an attractive benchmark for data-driven surrogate models.

The phase-field method has emerged as a powerful variational framework for simulating fracture evolution in solids [3, 4]. Instead of representing cracks as discrete discontinuities, the method introduces a continuous scalar damage field that smoothly transitions between intact and fractured material states. This formulation captures crack nucleation, deflection, arrest, and arbitrary propagation paths without any ad hoc tracking algorithm [5, 6]. The price of this generality is computational effort: resolving the diffuse crack zone requires an element size well below the regularization length of the crack, so each simulation of a heterogeneous specimen involves nearly $\mathcal{O}(10^5)$ degrees of freedom solved over hundreds of load increments. Generating a dataset large enough to train a surrogate model is consequently an expensive, resource-intensive undertaking.

At the same time, recent advances in machine learning, specifically neural operators, have created new opportunities for accelerated fracture prediction and surrogate modeling [7, 8, 9, 10]. Data-driven frameworks require high-quality benchmark datasets that span a well-characterized input space and expose the full response history rather than a single summary quantity. In this work, we present a dataset of high-fidelity phase-field simulations covering 1100 distinct configurations of a three-dimensional notched heterogeneous brittle-composite specimen, each defined by the position of the initial notch and the arrangement of four aggregate inclusions along its path. For every realization, the full displacement and phase-field histories are stored at 401 output frames (400 load increments) on the original finite element mesh, together with the low-dimensional parameters that generated them. The overall framework for dataset generation and machine learning integration is illustrated in Figure 1. The pipeline is organized into four sequential modules: Module 1 defines the fracture problem, encompassing geometry, the two-phase mesostructure, material properties, boundary conditions, loading, and the phase-field model; Module 2A handles dataset generation by systematically varying the notch height and the aggregate positions, computing the resulting displacement and damage fields, and storing all outputs at each load step using Module 1; Module 2B performs preprocessing, including extraction of the nodal fields, assembly into a compact self-contained archive, and preparation of the ML-ready inputs; and Module 3 covers the machine learning pipeline, from defining the learning task and model architecture to training, validation, and deployment for fast predictions. In this document we focus on Modules 1, 2A and 2B, i.e. the problem formulation, the generation

of the high-fidelity simulation data, and the packaging of that data for learning. The choice of machine learning model (Module 3) is intentionally left open, as it will depend on the specific surrogate modeling task adopted by the end user.

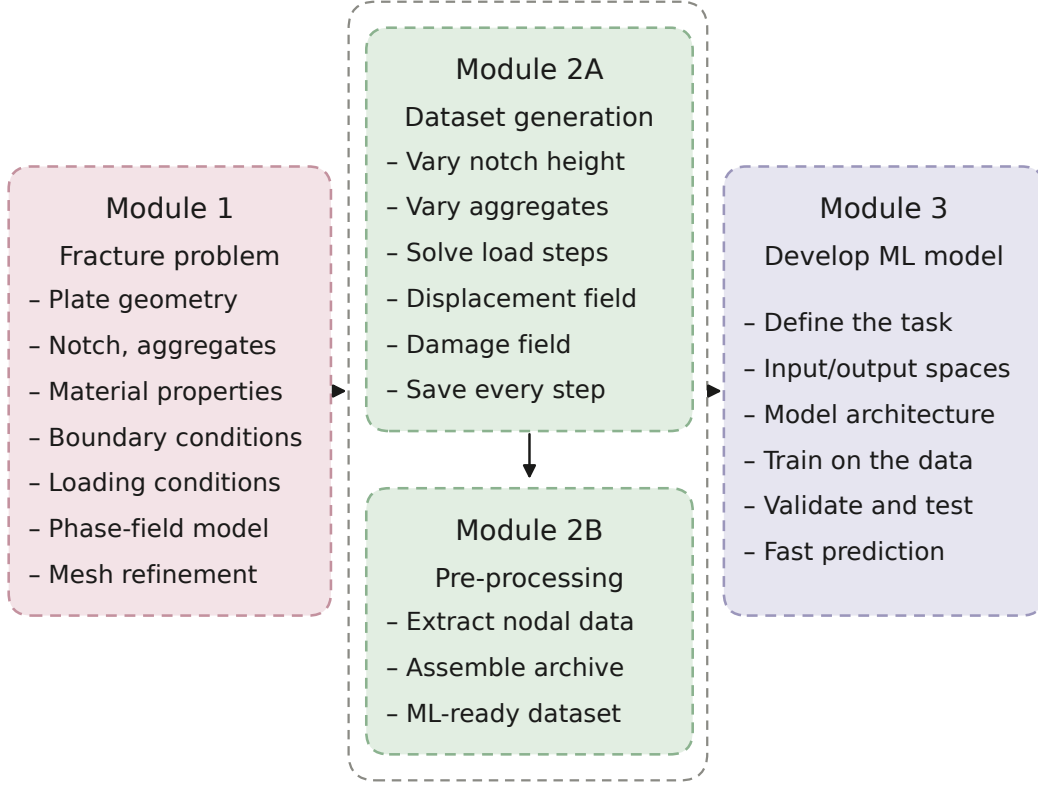


Figure 1: Overview of the computational workflow used to generate the heterogeneous phase-field fracture dataset. The workflow is organized into four modules: Module 1, definition of the fracture problem; Module 2A, systematic generation of the simulation data; Module 2B, pre-processing into an ML-ready archive; and Module 3, development of data-driven models for fracture prediction. Each realization is defined by the notch height and the longitudinal positions of four aggregate inclusions. The simulation advances through 400 load increments and stores the displacement and phase-field fields at 401 output frames.

2. Phase-Field fracture formulation

The dataset is based on the standard second-order (AT2) phase-field formulation for brittle fracture under quasi-static loading. Fracture is represented by a scalar damage variable $d \in [0, 1]$, where $d = 0$ denotes intact material and $d = 1$ indicates complete fracture. The total energy, $E(\mathbf{u}, d)$ of the system is expressed as the functional:

$$E(\mathbf{u}, d) = \int_{\Omega} \psi_s(\boldsymbol{\varepsilon}, d) dV + \int_{\Omega} \frac{G_c}{c_\alpha} \left[\frac{\alpha(d)}{\ell} + \ell |\nabla d|^2 \right] dV - P(\mathbf{u}), \quad (1)$$

where $\mathbf{u}$ denotes the vector-valued displacement field and $\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^\top)$ the infinitesimal strain. The boundary $\partial\Omega$ is partitioned into complementary parts Γ_u and Γ_t , on which the displacement and the traction are prescribed respectively. At each load step the solution is the minimizer of (1) over the admissible sets

$$\mathcal{U} = \{ \mathbf{u} \in [H^1(\Omega)]^3 : \mathbf{u} = \bar{\mathbf{u}} \text{ on } \Gamma_u \}, \quad \mathcal{D} = \{ d \in H^1(\Omega) : 0 \leq d \leq 1 \}, \quad (2)$$

where $\bar{\mathbf{u}}$ is the prescribed displacement on Γ_u ; irreversibility of damage is enforced through the history field introduced in (6). Here, ℓ is the internal length-scale parameter that controls the width of the diffused crack zone, G_c is the critical energy release rate, and $\alpha(d) = d^2$ is the crack geometry function for the AT2 model.

The scaling factor $c_\alpha := 4 \int_0^1 \sqrt{\alpha(\beta)} d\beta$ ($c_\alpha = 2$ for the AT2 model) maintains consistency in the crack density formulation. The external work is expressed:

$$P(\mathbf{u}) = \int_{\Omega} \mathbf{b} \cdot \mathbf{u} dV + \int_{\Gamma_t} \mathbf{t} \cdot \mathbf{u} dA \quad (3)$$

with $\mathbf{b}$ the body force and $\mathbf{t}$ the applied traction on the Neumann boundary Γ_t . Loading in the present dataset is quasi-static and displacement-controlled, so inertial contributions are omitted and the load is advanced through a sequence of equilibrium states. The strain energy density is defined as

$$\psi_s(\boldsymbol{\varepsilon}, d) = [\omega(d) + k] \psi_0(\boldsymbol{\varepsilon}), \quad \psi_0(\boldsymbol{\varepsilon}) = \frac{\lambda}{2} (\text{tr } \boldsymbol{\varepsilon})^2 + \mu \boldsymbol{\varepsilon} : \boldsymbol{\varepsilon}, \quad (4)$$

where $\omega(d) = (1 - d)^2$ is the degradation function satisfying $\omega(0) = 1$ and $\omega(1) = 0$, $k = 10^{-8}$ is a small residual stiffness that keeps the mechanical problem well posed in fully broken regions, and λ and μ are the Lamé parameters. A hybrid formulation is adopted [6]. The mechanical equilibrium problem uses the isotropically degraded stress, whereas the crack-driving force is computed from the tensile part of the strain energy. Following the spectral decomposition proposed by Miehe *et al.* [11], the tensile contribution is

$$\psi_0^+(\boldsymbol{\varepsilon}) = \frac{\lambda}{2} \langle \text{tr } \boldsymbol{\varepsilon} \rangle_+^2 + \mu (\boldsymbol{\varepsilon}^+ : \boldsymbol{\varepsilon}^+), \quad (5)$$

where $\langle \cdot \rangle_+$ denotes the positive part and $\boldsymbol{\varepsilon}^+$ is assembled from the positive parts of the principal strains, evaluated through a closed-form spectral decomposition of the 3×3 strain tensor. Irreversibility of damage is enforced through the history field

$$\mathcal{H}(\mathbf{x}, t) = \max_{s \leq t} \psi_0^+(\boldsymbol{\varepsilon}(\mathbf{u}(\mathbf{x}, s))), \quad (6)$$

which is monotonically non-decreasing in time and therefore prevents cracks from healing as the load increases. Using the history field in the crack-driving force, the hybrid formulation leads to the following coupled equilibrium equations, which are solved at each load increment:

$$\nabla \cdot \left([(1 - d)^2 + k] \boldsymbol{\sigma}_0(\mathbf{u}) \right) = 0, \quad \left(\frac{G_c}{\ell} + 2\mathcal{H} \right) d - G_c \ell \Delta d = 2\mathcal{H}, \quad (7)$$

where $\boldsymbol{\sigma}_0(\mathbf{u}) = 2\mu(\mathbf{x}) \boldsymbol{\varepsilon}(\mathbf{u}) + \lambda(\mathbf{x}) \text{tr } \boldsymbol{\varepsilon}(\mathbf{u}) \mathbf{I}$ is the undamaged elastic stress, with the Lamé parameters, $\lambda(\mathbf{x})$ and $\mu(\mathbf{x})$ determined from the spatially varying $E(\mathbf{x})$ and $\nu(\mathbf{x})$ in Eq. (9). The system is closed by the boundary conditions

$$\mathbf{u} = \bar{\mathbf{u}} \text{ on } \Gamma_u, \quad [(1 - d)^2 + k] \boldsymbol{\sigma}(\mathbf{u}) \cdot \mathbf{n} = \mathbf{t} \text{ on } \Gamma_t, \quad \nabla d \cdot \mathbf{n} = 0 \text{ on } \partial\Omega, \quad (8)$$

with $\mathbf{n}$ the outward unit normal. The homogeneous Neumann condition on the phase field is the natural boundary condition of (1) and applies on the whole boundary, including the notch faces. The specific choice of Γ_u , Γ_t and $\bar{\mathbf{u}}$ used to generate the dataset is given in the next section. This variational structure guarantees energy balance and thermodynamic consistency throughout the fracture process. The geometry, mesostructure, and boundary condition setup used in the simulations are shown in Figure 2.

3. Computational framework

The computational setup involves a coupled displacement and phase-field formulation for a three-dimensional single-edge-notched plate of dimensions $L_x \times L_y \times L_z = 1.0 \times 1.0 \times 0.1$ containing a pre-existing horizontal notch, as shown in Figure 2. The notch is introduced as a through-crack on the left face: its two faces are cut at $y = h \pm w_0/2$ on the plane $x = 0$ and converge to a single tip at (a, h) , where h is the notch height, $a = 0.1$ the notch length, and $w_0 = 10^{-3}$ the width of the notch opening at the left face. Since $w_0 = \ell/20$, the notch is far thinner than the diffuse crack zone and therefore acts as a sharp pre-existing crack rather than a blunt geometric

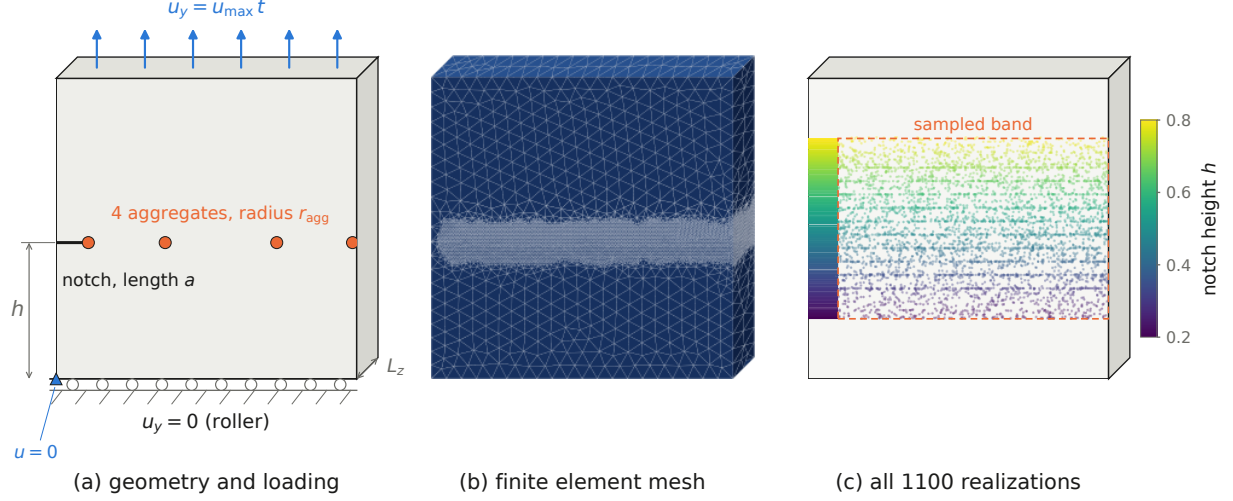


Figure 2: (a) Geometry and boundary condition setup for the concrete fracture simulations. A plate of dimensions $1.0 \times 1.0 \times 0.1$ contains a pre-existing horizontal notch of length a at height h , together with four stiff aggregate inclusions centered on the ligament ahead of the notch tip. A monotonically increasing tensile displacement is applied on the top boundary. (b) Tetrahedral finite element mesh of a representative realization, shown in the same oblique view as (a); the refined band that follows the sampled notch height and the expected crack path is clearly visible, and the mesh is regenerated for every realization. (c) All 1100 realizations at once, on the same plate and in the same view: the 4400 aggregate centres, and on the left face the notch trace of every realization, both coloured by notch height. The dashed box is the region the inputs are drawn from, $x_i \in [0.1, 1]$ and $h \in [0.2, 0.8]$; the plate is drawn in full so the unsampled margins are visible.

feature.

The specimen is modeled as a synthetic two-phase brittle composite. A homogeneous matrix phase is seeded with $N_{\text{agg}} = 4$ stiffer aggregate inclusions, each a circular disk of radius $r_{\text{agg}} = 0.02$ in the x - y plane extruded through the thickness, with center (x_i, h) for $i = 1, \dots, 4$. All four aggregate centers lie on the line $y = h$, that is, directly on the ligament ahead of the notch tip, so that the propagating crack is forced to interact with them. The material properties are assigned pointwise as

$$(E, \nu, G_c)(\mathbf{x}) = \begin{cases} (2E_m, 0.9\nu_m, 1.5G_{c,m}), & \text{if } \exists i : (x - x_i)^2 + (y - h)^2 \leq r_{\text{agg}}^2, \\ (E_m, \nu_m, G_{c,m}), & \text{otherwise,} \end{cases} \quad (9)$$

so that the aggregates are twice as stiff and 50% tougher than the surrounding matrix. Here $\mathbf{x}$ denotes the general position vector. Because the aggregate inclusions are circular disks extruded uniformly through the thickness, the material heterogeneity varies only in the x - y plane, while the displacement and phase-field response is solved in the full three-dimensional domain. The matrix parameters are held fixed across all simulations and are provided in Table 1 for reproducibility; they follow the parameter set widely used for the single-edge-notched tension benchmark [11].

Table 1: Material and model parameters used in all concrete fracture simulations. The matrix values are held fixed across the dataset; the aggregate values follow from Eq. (9). Note that the matrix properties follow the classical parameter set of the single-edge-notched tension benchmark [11] rather than experimentally calibrated concrete properties; they are adopted here to keep the baseline fracture response verifiable against this reference solution, while the aggregate-to-matrix contrast in Eq. (9) governs the mesoscale behavior explored in the dataset.

Phase	E [MPa]	ν [-]	G_c [N/mm]	ℓ [mm]	k [-]
Matrix	210×10^3	0.30	2.7	0.02	10^{-8}
Aggregate	420×10^3	0.27	4.05	0.02	10^{-8}

Loading is displacement-controlled and monotonic in tension (mode I). Denoting by $\mathbf{u} = (u_x, u_y, u_z)$ the displacement field, the prescribed conditions are $u_y = 0$ on $y = 0$, $u_y = u_{\max} t$ on $y = L_y$, and $u = 0$ on the edge $x = y = 0$. The bottom face is constrained only in the loading direction, and the additional pointwise constraint on the edge removes the remaining rigid-body translations. The two faces normal to z are traction-free. The pseudo-time t is advanced from 0 to 1 in 400 uniform increments, and the solution is written after every increment, giving 401 output frames per simulation.

Mesheres are generated with Gmsh [12]: the surfaces above and below the notch are meshed separately and extruded through the thickness. Resolving the diffuse crack requires elements finer than the regularization length ℓ , which is too expensive to afford over the whole specimen, so the element size is graded: inside a rectangular band enclosing the ligament ahead of the notch, where the crack is expected to run, the target element size is $h_{\min} = \ell/3$, while outside it the mesh is coarsened to $h_{\max} = 0.05$. Because that band follows the sampled notch height, the mesh is regenerated for every realization and its size varies across the dataset (Table 2); node counts and connectivity are therefore simulation-specific and must not be assumed identical across samples. The variational problems are discretized in FEniCS [13] with piecewise-linear elements for both $\mathbf{u}$ and d and piecewise-constant elements for the history field $\mathcal{H}$, and Equations (7) are solved with a single staggered pass per load increment.

The variability in the dataset is introduced through two factors: the notch height h , which locates the initial defect within the specimen, and the four aggregate abscissae $x_1, \dots, x_4$, which determine the mesostructure along the crack path. The notch height is sampled over $h \in [0.2, 0.8]$ and the aggregate abscissae uniformly on the ligament, $x_i \sim \mathcal{U}(a, 1)$ with $a = 0.1$. Everything else — the matrix and aggregate material properties, the notch length, the plate dimensions, the regularization length, the number of aggregates, and the loading protocol — is held fixed, so that the dataset isolates the effect of the mesostructure. This produces a total of **1100 simulations**, each described by the five scalars (x_1, x_2, x_3, x_4, h) . Table 2 summarizes the complete parameter space. Figure 3 presents representative phase-field results for realizations with contrasting aggregate arrangements, following each crack from initiation to full separation at equal fractions of the ligament crossed. Even though only five scalar parameters vary between simulations, the resulting fracture responses are highly diverse. Because the aggregate centers lie along the nominal ligament, changes in their longitudinal positions alter the spacing between inclusions and the interaction between the advancing crack and the heterogeneous material. The crack may propagate approximately straight through the ligament, deflect around an inclusion, or become temporarily arrested before resuming propagation. This range of behaviors underscores the diversity of the dataset.

Table 2: Parameter space of the dataset. Only the notch height and the four aggregate abscissae vary between simulations; all material and loading parameters are identical throughout.

Quantity	Symbol	Value
Plate dimensions	$L_x \times L_y \times L_z$	$1.0 \times 1.0 \times 0.1$
Notch length	a	0.1 (fixed)
Notch height	h	$[0.2, 0.8]$ (varies)
Notch opening width	w_0	10^{-3}
Number of aggregates	N_{agg}	4
Aggregate radius	r_{agg}	0.02
Aggregate centers	(x_i, h)	$x_i \sim \mathcal{U}(0.1, 1)$ (varies)
Maximum applied displacement	$u_{\max}$	0.02
Load increments / output frames	—	400 / 401
Nodes per mesh	N	42,406 – 44,690 (varies)
Tetrahedra per mesh	M	244,218 – 247,874 (varies)
Number of simulations	—	1100
Training / test split	—	880 / 220

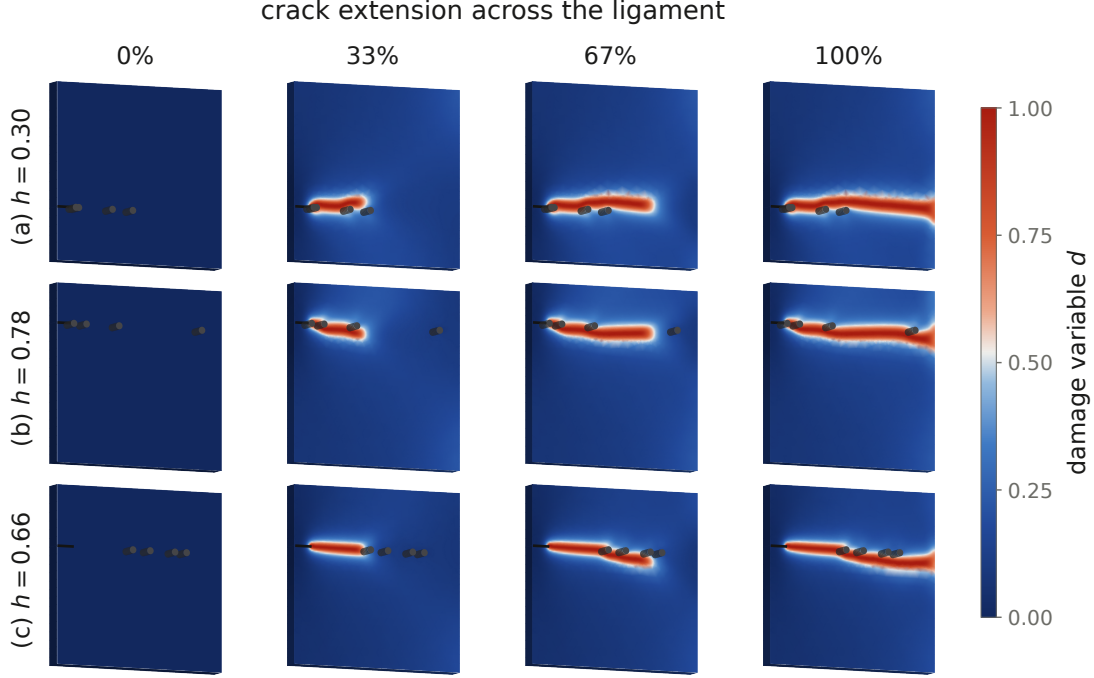


Figure 3: Damage evolution for three realizations (rows) with different aggregate arrangements under identical loading, rendered on the three-dimensional specimen. Columns correspond to equal fractions of the ligament crossed by the crack tip, selected separately for each realization because the fronts advance at different rates; the load step behind each column is therefore not the same across rows. The dark slit on the left face is the notch and the dark cylinders are the aggregate inclusions, which run through the full thickness. Depending on whether the aggregates lie on or off the ligament, the crack propagates straight through, deflects around an inclusion, or is temporarily arrested before resuming, demonstrating the range of fracture behaviors captured in the dataset.

4. Dataset description

The dataset is released as a single self-contained HDF5 archive holding the full simulation output, accompanied by three small companion files that describe the inputs, the train/test partition, and a derived summary of the final crack geometry. The organization is shown in Figure 4. Fields are stored in single precision with gzip compression, and the mesh of each simulation is stored once rather than repeated at every load step, giving a total size of approximately 270 GB for the 1100 simulations. No resampling, interpolation, or cropping is applied: the fields are provided at the original finite element nodes, exactly as computed by the solver, so that users are free to choose their own preprocessing strategy. The released files are the following:

- `processed_xyz_disp_joint1100_standalone.h5`: the main data file, containing 1100 groups named `job_0000 ... job_1099`, one per simulation. Each group holds the datasets listed in Table 3. Group attributes record the number of points and time steps, the channel names, the observed range of the phase field, and provenance fields identifying the generation batch.
- `branch5_inputs_joint1100.npz`: the low-dimensional input description for operator learning. It stores `job_names (1100,)`, `branch_inputs (1100, 5)`, `branch_order = [aggregate_x_0, ..., aggregate_x_3, initial_height]`, and `trunk_t (401, 1)`. The five branch scalars are exactly the quantities that vary across the dataset, and `trunk_t` is the normalized pseudo-time $t \in [0, 1]$ sampled at the 401 output frames.
- `split880_220_seed42.npz`: the training/test split, stored as `train_job_names (880,)` and `test_job_names (220,)` together with `all_job_names`, `seed = 42` and `train_ratio = 0.8`. The split is stratified by generation batch and is performed at the level of simulations, never of individual time frames, so that no test trajectory is partially seen during training.
- `true_endpointclean_centerline_joint1100.npz`: a derived, reduced description of the final crack

geometry. For each simulation the crack centerline is extracted from the final phase field by thresholding at $d = 0.95$ and resampled onto a fixed grid of 201 abscissae spanning $x \in [0, 1]$, with spurious endpoints removed. It stores `job_names` (1100,), `x_grid` (201,) and `centerline` (1100, 201). This file is provided as a convenience for crack-path prediction tasks and can be regenerated from the phase field.

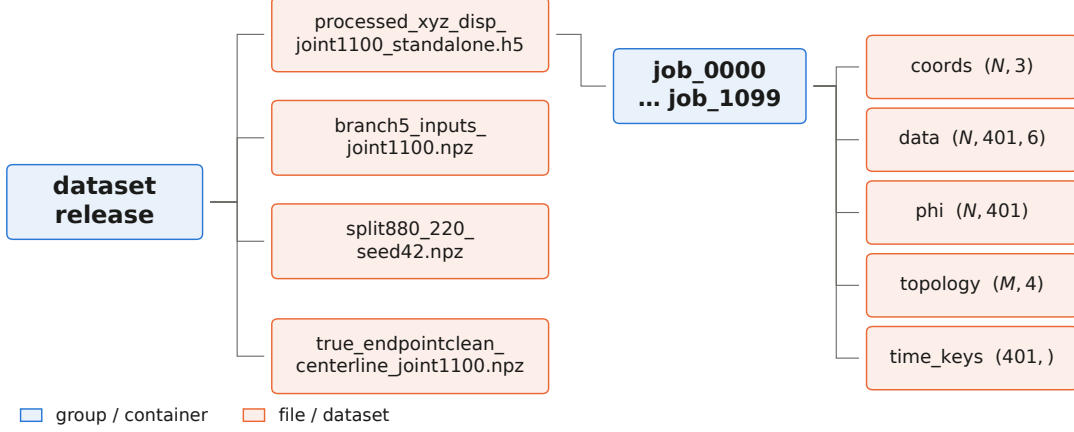


Figure 4: Structure of the released dataset: a single self-contained HDF5 archive holding the displacement and phase-field histories of all 1100 simulations, together with the companion files describing the simulation inputs, the training/test split, and the extracted crack centerlines.

Three conventions are worth emphasizing when using the archive. First, the first three channels of `data` are the *reference* coordinates and are constant in time; they are replicated along the time axis purely for convenience and are identical to `coords`. The deformed configuration is obtained as $x + u$. Second, because the mesh is regenerated for every realization, node j of one simulation has no correspondence with node j of another, and node counts differ between simulations. Any model that batches multiple simulations together must therefore handle variable node counts, either through padding and masking or through an architecture that is invariant to the number and ordering of points. By contrast, the temporal discretization *is* shared: every simulation contains the complete sequence of 401 frames under an identical loading protocol, so no masking is required along the time axis and the trunk input is common to all samples. Third, the mesostructure of a given simulation is fully specified by the five scalars in `branch5_inputs_joint1100.npz` together with the fixed parameters of Table 1: the aggregate regions can be reconstructed on any mesh by evaluating the indicator in Eq. (9), so the material fields need not be stored per node. Figure 5 shows how the three displacement components evolve over the loading history for a representative realization.

Table 3: Contents of each simulation group in `processed_xyz_disp_joint1100_standalone.h5`. Here N is the number of nodes and M the number of tetrahedra of that particular simulation; both are simulation-specific.

Dataset	Shape and type	Description
<code>coords</code>	$(N, 3)$, float32	reference (undeformed) nodal coordinates
<code>data</code>	$(N, 401, 6)$, float32	channels $[x, y, z, u_x, u_y, u_z]$
<code>phi</code>	$(N, 401)$, float32	phase-field damage $d \in [0, 1]$
<code>topology</code>	$(M, 4)$, int32	tetrahedral connectivity
<code>time_keys</code>	$(401,)$, int32	solver frame index of each load step

The archive contains the two nodal fields required to reconstruct the mechanical and damage state at every load step, namely the displacement vector and the phase field. The cell-wise crack driving force $\mathcal{H}$ of Eq. (6) is not distributed, as it is a derived internal variable that can be recomputed from the stored displacement history through Eqs. (5) and (6).

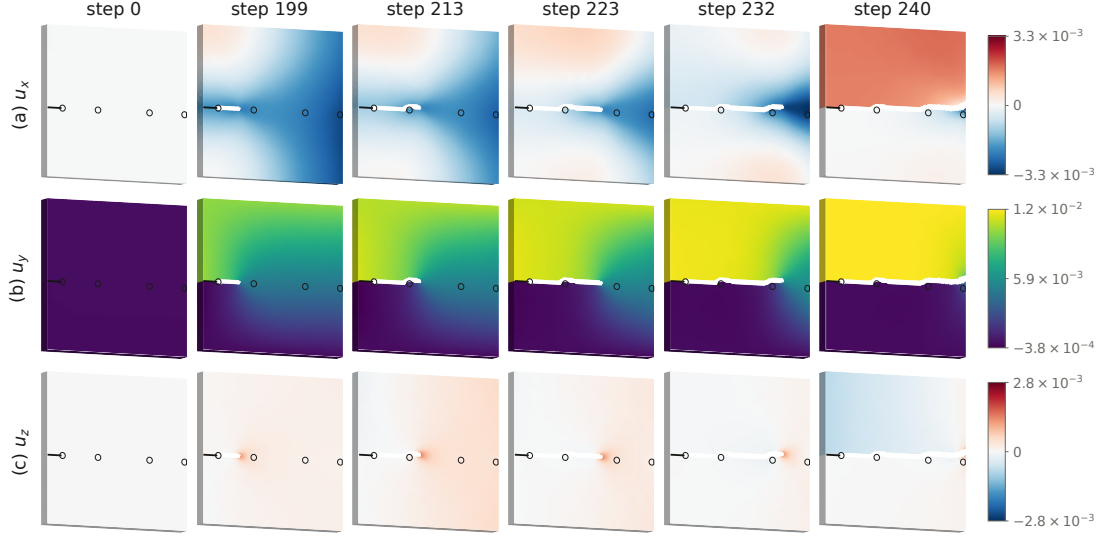


Figure 5: Evolution of the three displacement components for the realization of Figure 2(a,b), rendered on the three-dimensional specimen in the same view as Figure 3. Each row is one component (u_x , u_y , u_z) and each column one load step, with the steps chosen so that the crack tip advances by roughly equal increments between columns. Within a row the color scale is common to all steps, so the columns are directly comparable; between rows it is not, since the three components differ by more than an order of magnitude. Aggregate outlines are overlaid in black. The thin white band that opens along the ligament marks the fully damaged elements, which are removed rather than interpolated through because the displacement is discontinuous across an open crack. The dominant response is the mode-I opening in u_y , in which the two halves of the specimen separate almost rigidly; u_x concentrates around the advancing crack tip, and u_z reflects the through-thickness Poisson contraction.

5. Community use

This dataset is openly released to support the development, training, and benchmarking of data-driven models for heterogeneous fracture. The dataset is particularly suited to methods that learn mappings from a low-dimensional description of the specimen configuration to the evolving displacement and phase-field response on simulation-specific unstructured meshes. Potential applications include surrogate modeling and neural operators, crack-path prediction, quantification of inclusion-induced crack deflection and arrest, uncertainty quantification, and assessment of model generalization to aggregate configurations not encountered during training. By releasing this dataset, we aim to lower the barrier to entry for researchers working at the intersection of computational mechanics and machine learning, and to promote reproducible, community-driven advances in the simulation and prediction of fracture in concrete and other heterogeneous quasi-brittle materials.

References

- [1] A. A. Griffith. The phenomena of rupture and flow in solids. *Philosophical Transactions of the Royal Society London A* 221, pages 163–198, 1921.
- [2] G. R. Irwin. Fracture. *Elasticity and Plasticity, Encyclopedia of Physics, Springer*, pages 551–590, 1958.
- [3] Gilles A Francfort and Jean-Jacques Marigo. Revisiting brittle fracture as an energy minimization problem. *Journal of the Mechanics and Physics of Solids*, 46(8):1319–1342, 1998.
- [4] B. Bourdin, G. A. Francfort, and J. J. Marigo. Numerical experiments in revisited brittle fracture. *Journal of the Mechanics and Physics of Solids*, 48(4):797–826, 2000.
- [5] Christian Miehe, Fabian Welschinger, and Martina Hofacker. Thermodynamically consistent phase-field models of fracture: Variational principles and multi-field FE implementations. *International journal for numerical methods in engineering*, 83(10):1273–1311, 2010.

- [6] Marreddy Ambati, Tymofiy Gerasimov, and Laura De Lorenzis. A review on phase-field models of brittle fracture and a new fast hybrid formulation. *Computational Mechanics*, 55(2):383–405, 2015.
- [7] Somdatta Goswami, Aniruddha Bora, Yue Yu, and George Em Karniadakis. Physics-informed deep neural operator networks. In *Machine learning in modeling and simulation: methods and applications*, pages 219–254. Springer, 2023.
- [8] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. *arXiv preprint arXiv:2010.08895*, 2020.
- [9] Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via deepoNet based on the universal approximation theorem of operators. *Nature machine intelligence*, 3(3):218–229, 2021.
- [10] Somdatta Goswami, Minglang Yin, Yue Yu, and George Em Karniadakis. A physics-informed variational DeepONet for predicting crack path in quasi-brittle materials. *Computer Methods in Applied Mechanics and Engineering*, 391:114587, 2022.
- [11] Christian Miehe, Martina Hofacker, and Fabian Welschinger. A phase field model for rate-independent crack propagation: Robust algorithmic implementation based on operator splits. *Computer Methods in Applied Mechanics and Engineering*, 199(45-48):2765–2778, 2010.
- [12] Christophe Geuzaine and Jean-François Remacle. Gmsh: A 3-D finite element mesh generator with built-in pre- and post-processing facilities. *International Journal for Numerical Methods in Engineering*, 79(11):1309–1331, 2009.
- [13] Martin Alnæs, Jan Blechta, Johan Hake, August Johansson, Benjamin Kehlet, Anders Logg, Chris Richardson, Johannes Ring, Marie E. Rognes, and Garth N. Wells. The FEniCS project version 1.5. *Archive of Numerical Software*, 3(100):9–23, 2015.